MA 5755: Data Analysis & Visualization in R/Python/SQL

Week 4: Probabilistic Clustering and Gaussian Mixture Models

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Outline

1. Review
2. Computational Cost
3. Mixture Models for Density Estimation and Classification
4. The EM Algorithm
5. EM Algorithm for Gaussian Mixture Models
6. Probabilistic Clustering

Distance Determines the Method

Many learning algorithms depend entirely on a distance $d(x_i, x_j)$.

k-means objective

$$\sum_{k=1}^{K} \sum_{i \in \mathcal{C}_k} \|x_i - \mu_k\|_2^2.$$

k-medoids objective

$$\sum_{k=1}^{K} \sum_{i \in \mathcal{C}_k} d(x_i, m_k), \quad m_k \in \mathcal{C}_k.$$

Key differences:

- k-means minimizes squared Euclidean distortion.
- k-medoids minimizes unsquared distance under a chosen metric.
- k-means representatives are means.
- k-medoids representatives are observed data points.

The representative is determined by the loss induced by the distance.

Implications for kNN and k-Means

kNN.
Prediction at x_0 is $\hat{g}(x_0) =$ majority vote among $\{x_{(1)}, \ldots, x_{(k)}\}$.
As the dimension increases:

- Neighborhood radius grows.
- "Local" neighborhoods include distant points.
- Averaging becomes less localized.

k-means.
Objective is $\sum_{k=1}^{K} \sum_{i \in \mathcal{C}_k} \|x_i - \mu_k\|^2$.

- Assignments induce Voronoi regions
- If all distances are similar, boundaries become unstable.
- Small centroid changes can alter large regions.
- Clusters may reflect noise rather than structure.

When k-Means Works Well

k-means when geometric assumptions are approximately satisfied:

1. Approximately spherical clusters. If within-cluster covariance is roughly isotropic, $\mathrm{Cov}_k \approx \sigma^2 I$, the mean is a meaningful summary.

2. Comparable within-cluster variance. If one cluster has much larger spread, the algorithm may split it instead of separating distinct groups.

3. Informative dimensions.
If only $L \ll p$ coordinates contain signal, then noise dimensions inflate distortion and blur separation.

4. Proper scaling is mandatory because variables with larger scale dominate unless standardized.

k-medoids can be more robust when outliers are present, mean is not a meaningful representative, non-Euclidean distances are desired.

Structural Properties and Practical Implications

Optimization structure.

k-means: k-means alternates between the assignment step and the update step (closed-form mean). Each step decreases the objective and convergence is guaranteed in finitely many steps, but only to a local minimum.

k-medoids:

- No closed-form update.
- Combinatorial search over candidate representatives.
- Typically more computationally expensive.

Takeaway.

Clustering methods approximate geometry by K representatives. Their success depends on how well the chosen distance reflects structure in the data.

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Computational Cost of Nearest-Neighbor Methods

Nearest-neighbor methods place most of their burden at prediction time.

For N training points in $\mathbb{R}^p$, a single kNN query requires:

$$O(Np)$$

operations to compute distances and identify neighbors.

- The full training set must be stored.
- Prediction cost grows linearly with both N and p.
- This contrasts with parametric models, where prediction is $O(p)$.

Various data structures (e.g., k-d trees, ball trees) can reduce average search cost, but their effectiveness deteriorates as dimension increases.

Editing and Condensing the Training Set

Reducing storage and computation motivates pruning the training data.

Only a subset of training points is needed for accurate nearest-neighbor prediction, especially those near decision boundaries.

- **Editing methods:** remove points that are consistently misclassified.
- **Condensing methods:** keep only points essential for defining class boundaries.

Examples.

- **Multi-edit algorithm (Devijver & Kittler, 1982):** iteratively deletes misclassified points using cyclic train–test splits.
- **Condensed NN (Hart, 1968):** incrementally builds a minimal subset by adding points only when needed to correct misclassification.

These methods can reduce storage substantially, but lack strong guarantees and have seen limited systematic evaluation.

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Gaussian Mixture Model

Mixture density with K Gaussian components:

$$f(x) = \sum_{k=1}^{K} \pi_k \phi(x; \mu_k, \Sigma_k), \qquad \pi_k \geq 0, \qquad \sum_{k=1}^{K} \pi_k = 1.$$

Each component:

$$\phi(x; \mu_k, \Sigma_k) = \frac{1}{(2\pi)^{p/2} |\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right).$$

A Gaussian mixture model (GMM) is a flexible parametric density estimator.

General mixture models allow arbitrary component densities; Gaussian mixtures are the most common.

Maximum Likelihood Estimation

Given i.i.d. data $\{x_i\}_{i=1}^n$:

$$\ell(\theta) = \sum_{i=1}^{n} \log \left(\sum_{k=1}^{K} \pi_k \phi(x_i; \mu_k, \Sigma_k) \right).$$

Maximizing this log-likelihood is a non-convex optimization problem primarily due to the log of sums. We therefore use the Expectation-Maximization (EM) algorithm.

Posterior responsibilities (soft cluster memberships):

$$\gamma_{ik} = \frac{\pi_k \phi(x_i; \mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \phi(x_i; \mu_j, \Sigma_j)}.$$

Each γ_{ik} is the probability that observation i belongs to component k.

Spl. Case 1: Scalar Covariance $\Rightarrow$ Radial Form

Consider $\Sigma_k = \sigma_k^2 I$. Then

$$f(x) = \sum_{k=1}^{K} \pi_k \frac{1}{(2\pi\sigma_k^2)^{p/2}} \exp\left(-\frac{\|x-\mu_k\|^2}{2\sigma_k^2}\right).$$

Each component depends only on $\|x-\mu_k\|^2$, so it is **radial about** μ_k.

Define

$$K_k(x) = \frac{1}{(2\pi\sigma_k^2)^{p/2}} \exp\left(-\frac{\|x-\mu_k\|^2}{2\sigma_k^2}\right).$$

Then

$$f(x) = \sum_{k=1}^{K} \pi_k K_k(x) = \sum_{k=1}^{K} \pi_k K(\|x-\mu_k\|).$$

A Gaussian mixture with isotropic covariance is therefore equivalent to a **weighted radial basis expansion**.

Spl. Case 2: Limit to Kernel Density Estimation

Assume $\Sigma_k = \sigma^2 I$ with fixed $\sigma > 0$ and let $K \uparrow n$.

The MLE approaches:

$$\hat{\pi}_k = \frac{1}{n}, \qquad \hat{\mu}_k = x_k.$$

The resulting estimator becomes

$$\hat{f}(x) = \frac{1}{n} \sum_{k=1}^{n} \phi(x; x_k, \sigma^2 I).$$

This is precisely the **Gaussian kernel density estimator**.

Thus, kernel density estimation appears as a limiting case of GMM.

Heart Disease Risk-Factor Example

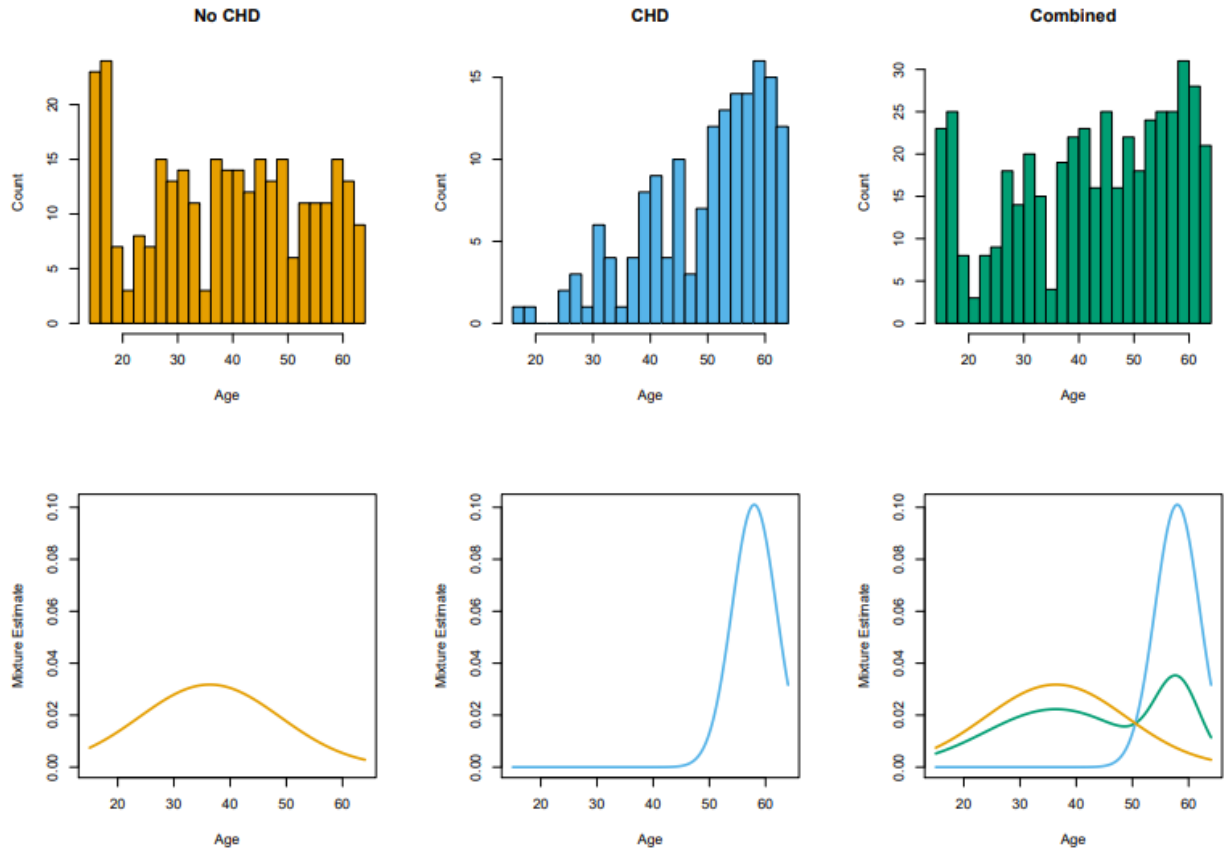

FIGURE 6.17. *Application of mixtures to the heart disease risk-factor study. (Top row:) Histograms of* `Age` *for the* `no CHD` *and* `CHD` *groups separately, and combined. (Bottom row:) estimated component densities from a Gaussian mixture model, (bottom left, bottom middle); (bottom right:) Estimated component densities (blue and orange) along with the estimated mixture density (green). The orange density has a very large standard deviation, and approximates a uniform density.*

Heart Disease Risk-Factor Example (contd.)

Typically, a large $\hat{\Sigma}_1$ approximates broad (near-uniform) density.

Estimated parameters: $\hat{\mu}_1 = 36.4, \hat{\Sigma}_1 = 157.7, \hat{\pi}_1 = 0.7$ and $\hat{\mu}_2 = 58.0, \hat{\Sigma}_2 = 15.6, \hat{\pi}_2 = 0.3$.

Fitting ignores CHD labels.

Using a threshold rule $\hat{\delta}_i = I(\hat{\gamma}_{i2} > 0.5)$, we get the following comparison with CHD labels:

	$\hat{\delta} = 0$	$\hat{\delta} = 1$
No CHD	232	70
CHD	76	84

Misclassification rate of about 32%, comparable to logistic regression.

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Two-Component Mixture Model

We model apparent bimodality using a two–Gaussian mixture.

$$Y_1 \sim N(\mu_1, \sigma_1^2), \qquad Y_2 \sim N(\mu_2, \sigma_2^2),$$

$$Y = (1-\Delta)Y_1 + \Delta Y_2, \qquad \Delta \in \{0,1\}, \qquad \Pr(\Delta = 1) = \pi.$$

The density is

$$g_Y(y) = (1-\pi)\phi_{\theta_1}(y) + \pi\phi_{\theta_2}(y), \tag{8.37}$$

with parameters

$$\theta = (\pi, \mu_1, \sigma_1^2, \mu_2, \sigma_2^2).$$

The latent variable Δ encodes component membership.

Observed vs Complete Likelihood

Observed log-likelihood:

$$\ell(\theta; Z) = \sum_{i=1}^{N} \log\left[(1-\pi)\phi_{\theta_1}(y_i) + \pi\phi_{\theta_2}(y_i)\right].$$

Maximization is difficult due to the sum inside the logarithm. Introduce latent indicators Δ_i. If $\Delta_i = 1$, then Y_i comes from model 2 and model 1 otherwise.

If Δ_i were known, the complete log-likelihood becomes

$$\ell_0(\theta; Z, \Delta) = \sum_{i=1}^{N} \Big[(1-\Delta_i)\log\phi_{\theta_1}(y_i) + \Delta_i \log\phi_{\theta_2}(y_i)\Big]$$
$$+ \sum_{i=1}^{N} \Big[(1-\Delta_i)\log(1-\pi) + \Delta_i \log\pi\Big].$$

Now maximization separates into weighted Gaussian fits.

EM algorithm

Since Δ_i's are unknown, we have to find the estimates iteratively. The γ_i's are called responsibilities.

Algorithm 8.2 *The EM Algorithm.*

1. Start with initial guesses for the parameters $\hat{\theta}^{(0)}$.
2. *Expectation Step*: at the jth step, compute

$$Q(\theta', \hat{\theta}^{(j)}) = \mathrm{E}(\ell_0(\theta'; \mathbf{T})|\mathbf{Z}, \hat{\theta}^{(j)}) \tag{8.43}$$

   as a function of the dummy argument θ'.
3. *Maximization Step*: determine the new estimate $\hat{\theta}^{(j+1)}$ as the maximizer of $Q(\theta', \hat{\theta}^{(j)})$ over θ'.
4. Iterate steps 2 and 3 until convergence.

Initialization and Local Maxima

Initialization

- Choose $\hat{\mu}_1, \hat{\mu}_2$ as two randomly selected y_i.
- Set $\hat{\sigma}_1^2 = \hat{\sigma}_2^2 = \frac{1}{N}\sum_{i=1}^{N}(y_i - \bar{y})^2$,
- Set $\hat{\pi} = 0.5$.

Likelihood behavior

Actual maximizer is possible when we put a spike of infinite height at any one data point, that is, $\hat{\mu}_k = y_i, \hat{\sigma}_k^2 = 0 \Rightarrow \ell(\theta; Z) \to \infty$.

But this is degenerate and not a very useful model. In practice, we search for a local maximum with $\hat{\sigma}_1^2, \hat{\sigma}_2^2 > 0$, noting multiple local maxima may exist.

Example

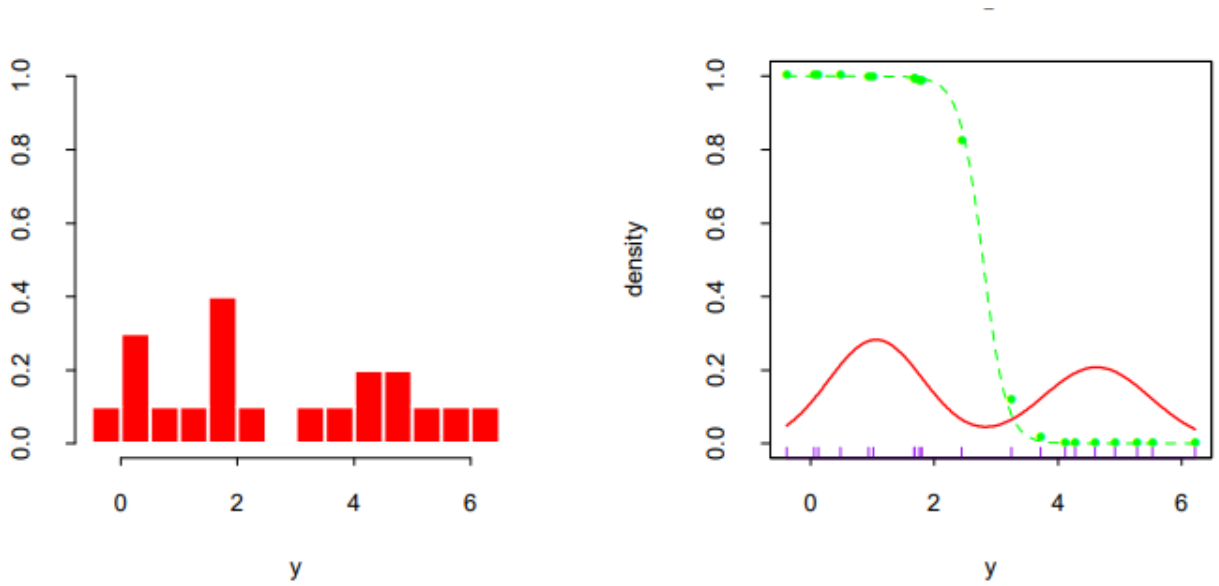

FIGURE 8.5. *Mixture example. (Left panel:) Histogram of data. (Right panel:) Maximum likelihood fit of Gaussian densities (solid red) and responsibility (dotted green) of the left component density for observation y, as a function of y.*

TABLE 8.1. *Twenty fictitious data points used in the two-component mixture example in Figure 8.5.*

-0.39	0.12	0.94	1.67	1.76	2.44	3.72	4.28	4.92	5.53
0.06	0.48	1.01	1.68	1.80	3.25	4.12	4.60	5.28	6.22

Example (contd.)

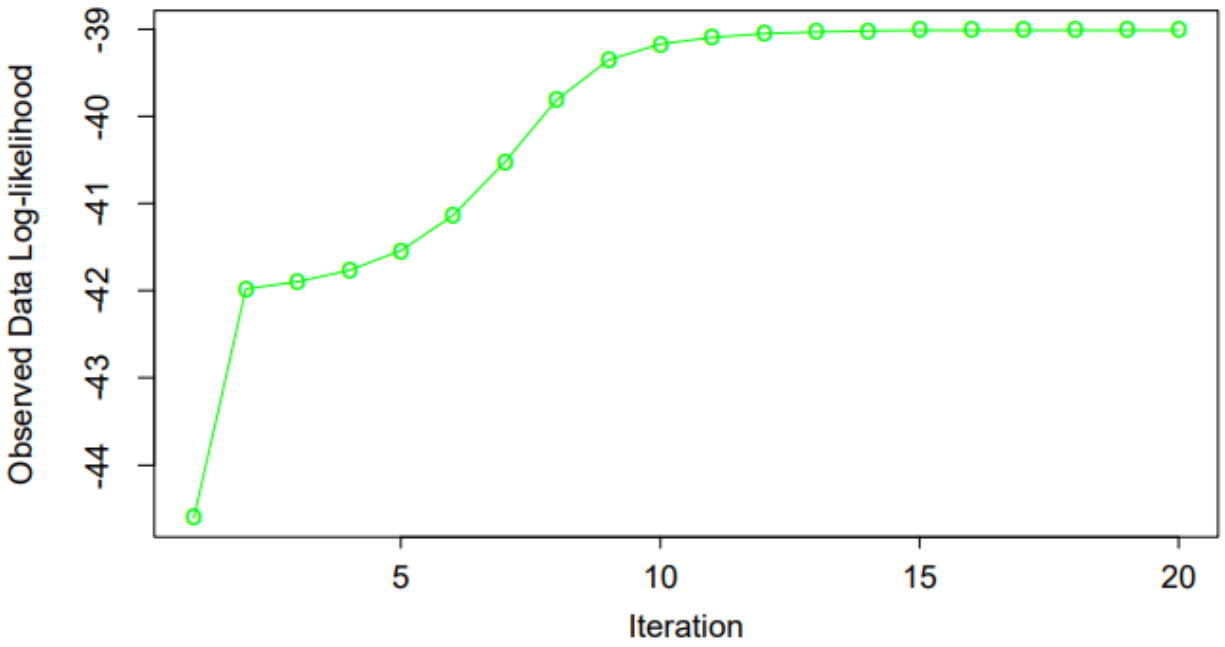

FIGURE 8.6. *EM algorithm: observed data log-likelihood as a function of the iteration number.*

Example (contd.)

TABLE 8.2. *Selected iterations of the EM algorithm for mixture example.*

Iteration	$\hat{\pi}$
1	0.485
5	0.493
10	0.523
15	0.544
20	0.546

The final maximum likelihood estimates are

$$\hat{\mu}_1 = 4.62, \qquad \hat{\sigma}_1^2 = 0.87,$$
$$\hat{\mu}_2 = 1.06, \qquad \hat{\sigma}_2^2 = 0.77,$$
$$\hat{\pi} = 0.546.$$

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Extension to K-Component GMM

We now extend the same EM principle to a general K-component Gaussian mixture in p dimensions.

Data: $x_1, \ldots, x_n$.

Latent indicators:

$$Z_i \in \{1, \ldots, K\}, \qquad P(Z_i = k) = \pi_k.$$

Conditional density:

$$p(x_i \mid Z_i = k) = \phi(x_i; \mu_k, \Sigma_k).$$

This is the direct generalization of the two-component model with $\Delta_i \in \{0, 1\}$.

Complete-Data Log-Likelihood (General Case)

As before, if the latent labels were observed, the likelihood would separate.

Define indicator variables:

$$z_{ik} = \mathbf{1}(Z_i = k).$$

The complete-data log-likelihood becomes

$$\ell_c(\theta) = \sum_{i=1}^{n} \sum_{k=1}^{K} z_{ik} \left[\log \pi_k + \log \phi(x_i; \mu_k, \Sigma_k) \right].$$

Exactly as in the two-component case, maximization now separates by component.

EM Algorithm for GMM

E-step: compute posterior responsibilities

$$\gamma_{ik} = P(Z_i = k \mid x_i; \theta^{(t)}) = \frac{\pi_k^{(t)} \phi(x_i; \mu_k^{(t)}, \Sigma_k^{(t)})}{\sum_{j=1}^{K} \pi_j^{(t)} \phi(x_i; \mu_j^{(t)}, \Sigma_j^{(t)})}.$$

M-step: maximize expected complete likelihood

$$N_k = \sum_{i=1}^{n} \gamma_{ik}, \qquad \pi_k^{(t+1)} = \frac{N_k}{n},$$

$$\mu_k^{(t+1)} = \frac{1}{N_k} \sum_{i=1}^{n} \gamma_{ik} x_i,$$

$$\Sigma_k^{(t+1)} = \frac{1}{N_k} \sum_{i=1}^{n} \gamma_{ik} (x_i - \mu_k^{(t+1)})(x_i - \mu_k^{(t+1)})^T.$$

Each iteration increases the observed log-likelihood.

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From Prototype-Based to Probabilistic Clustering

1. k-means (Prototype Method)

$$\min_{\{\mu_k\}} \sum_{i=1}^{n} \min_{k} \|x_i - \mu_k\|_2^2.$$

- Hard assignment of each x_i to exactly one cluster
- Objective: minimize within-cluster squared distances

2. Model-Based Clustering (GMM)

$$\max_{\theta} \sum_{i=1}^{n} \log \left(\sum_{k=1}^{K} \pi_k \phi(x_i; \mu_k, \Sigma_k) \right).$$

- Clusters correspond to mixture components
- Objective: maximize likelihood under a generative model

Conceptual Shift

k-means interpretation:
Each observation is **forced** into one cluster.

Probabilistic clustering:

- Each observation has posterior probabilities

$$\gamma_{ik} = P(Z_i = k \mid x_i).$$

- Clustering is inference on a latent variable.
- Assignments are **soft** rather than deterministic.

Core shift

- Distance minimization $\rightarrow$ likelihood maximization
- Deterministic labels $\rightarrow$ probabilistic membership
- Prototypes $\rightarrow$ generative components

Soft vs Hard Assignments

EM produces responsibilities:

$$\gamma_{ik} = P(Z_i = k \mid x_i).$$

These can be used in two ways:

- **Soft clustering:** retain full posterior probabilities
- **Hard clustering:**

$$\hat{z}_i = \arg\max_k \gamma_{ik}$$

Hard clustering from GMM is a **Bayes classification rule**.

An example

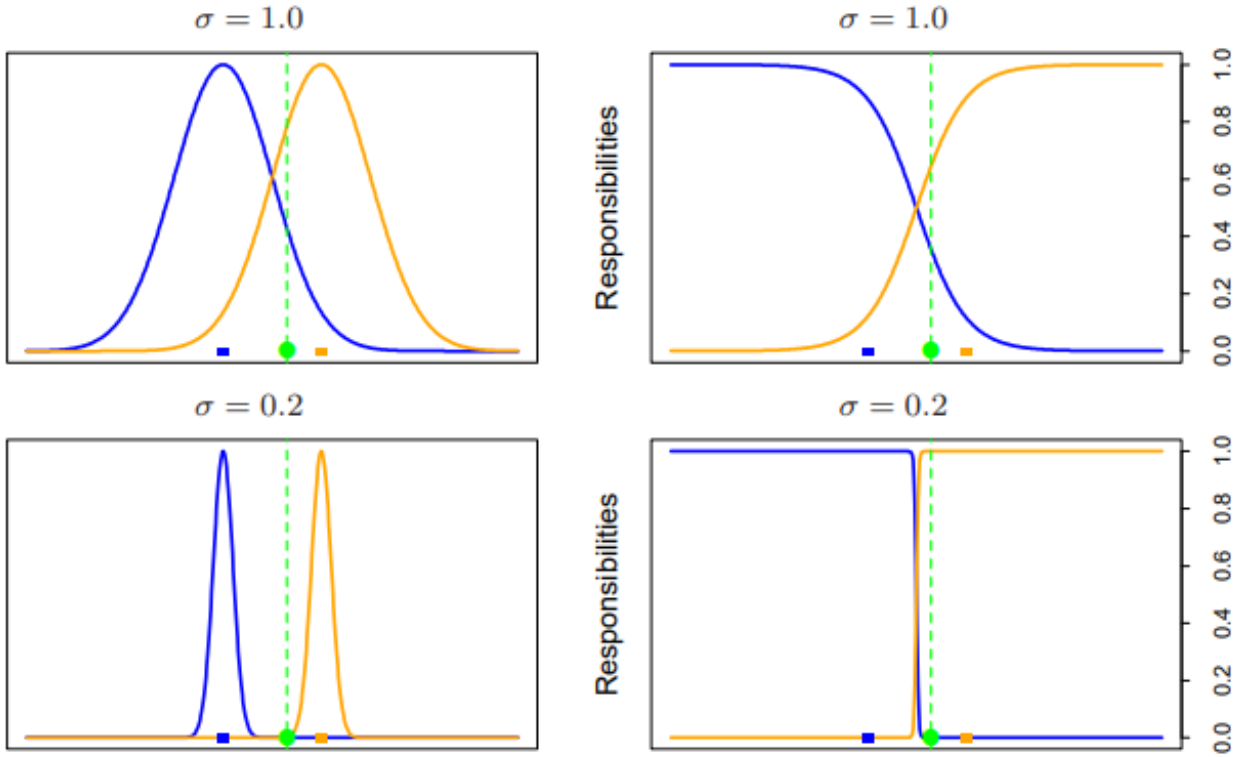

FIGURE 14.7. *(Left panels:) two Gaussian densities $g_0(x)$ and $g_1(x)$ (blue and orange) on the real line, and a single data point (green dot) at $x = 0.5$. The colored squares are plotted at $x = -1.0$ and $x = 1.0$, the means of each density. (Right panels:) the relative densities $g_0(x)/(g_0(x) + g_1(x))$ and $g_1(x)/(g_0(x) + g_1(x))$, called the "responsibilities" of each cluster, for this data point. In the top panels, the Gaussian standard deviation $\sigma = 1.0$; in the bottom panels $\sigma = 0.2$. The EM algorithm uses these responsibilities to make a "soft" assignment of each data point to each of the two clusters. When σ is fairly large, the responsibilities can be near 0.5 (they are 0.36 and 0.64 in the top right panel). As $\sigma \to 0$, the*

k-Means as Small-Variance Limit of GMM

Restricted GMM with $\Sigma_k = \sigma^2 I$ (spherical, equal variance) and $\pi_k = \frac{1}{K}$ (equal weights).

E-step reduces to

$$\gamma_{ik} \propto \exp\left(-\frac{1}{2\sigma^2}\|x_i - \mu_k\|^2\right).$$

Responsibilities depend only on squared distance.

As $\sigma^2 \to 0$

- Posterior concentrates on the nearest mean
- Soft assignments $\to$ hard assignments

$$\gamma_{ik} \to \mathbf{1}\left(k = \arg\min_j \|x_i - \mu_j\|^2\right)$$

Likelihood maximization reduces to the k-means objective.

Likelihood Pathology in GMM

Suppose for some component k:

$$\mu_k \to x_i, \qquad \Sigma_k \to 0.$$

Then the Gaussian density at x_i satisfies

$$\phi(x_i; \mu_k, \Sigma_k) \propto |\Sigma_k|^{-1/2} \to \infty.$$

Hence:

$$\ell(\theta) \to \infty.$$

Maximum likelihood is unbounded.

Consequences and Mitigation

Consequences

- Degenerate solutions (component collapses onto one point)
- Multiple local maxima
- Sensitivity to initialization

Mitigation

- Multiple random restarts
- Covariance regularization (e.g. minimum eigenvalues)
- Bayesian priors on parameters

Let us move to the (small) lab!